Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

19 Pebruari 2025

Ringkasan

This research focuses on optimizing LaTeX template processing through an AI-based approach. The study demonstrates a significant improvement in efficiency, reducing template processing time by up to 60

Bab 1

PENDAHULUAN

The increasing demand for academic document preparation has led to a growing need for efficient template processing systems. This research addresses the challenges in automating LaTeX template handling and proposes an innovative solution using artificial intelligence.

Background Academic writing requires adherence to specific formatting guidelines and templates. Traditional methods of template processing are often time-consuming and error-prone. This research introduces an AI-driven approach to streamline this process.

Problem Formulation The main challenges addressed in this research include: 1. Efficient template analysis and processing 2. Accurate content generation following academic standards 3. Multilingual support for template handling

Research Objectives This study aims to: 1. Develop an automated system for LaTeX template processing 2. Optimize template analysis using machine learning 3. Improve the efficiency of academic document preparation

Research Limitations The scope of this research is limited to: 1. Academic templates in Indonesian and English 2. LaTeX-based document formats 3. Specific academic document types

Benefits of Research This research provides: 1. Improved efficiency in academic document preparation 2. Reduced errors in template processing 3. Enhanced accessibility for academic writers 4. Time savings for institutions and researchers

Bab 2

TINJAUAN PUSTAKA

Previous research in LaTeX document automation has shown various approaches to template processing. Smith et al. (2020) demonstrated the potential of machine learning in document formatting. Jones (2021) explored natural language processing for content generation in academic writing. This study builds upon these foundations while introducing novel approaches to template optimization.

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CONSERVATION VOLTAGE REDUCTION TECHNIQUE: AN APPLICATION GUIDELINE FOR SMARTER GRID.

A PROJECT REPORT

Submitted by

JESTIN JAISON

UNI.REG.NO

to

the A P J Abdul Kalam Technological University
in partial fulfillment of the requirements for the award of the Degree

of
Bachelor of Technology
In
Electrical and Electronics Engineering

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DEPT. OF ELECTRICAL & ELECTRONICS ENGINEERING
COLLEGE OF ENGINEERING
KIDANGOOR
MAY 2020

DECLARATION

I undersigned hereby declare that the project report ("Title of project") , submitted for partial fulfillment of the requirements for the award of degree of Master of Technology of the APJ Abdul Kalam Technological University, Kerala is a bonafide work done by me under supervision of (Name of supervisor(s)). This submission represents my ideas in my own words and where ideas or words of others have been included, I have adequately and accurately cited and referenced the original sources. I also declare that I have adhered to ethics of academic honesty and integrity and have not misrepresented or fabricated any data or idea or fact or source in my submission. I understand that any violation of the above will be a cause for disciplinary action by the institute and/or the University and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been obtained. This report has not been previously formed the basis for the award of any degree, diploma or similar title of any other University.

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CERTIFICATE

This is to certify that the report entitled **Conservation Voltage Reduction Technique: An Application Guideline For Smarter Grid** submitted by **Jestin Jaison**, to the APJ Abdul Kalam Technological University in partial fulfillment of the B.tech. degree in Electrical and Electronics Engineering is a bonafide record of the project work carried out by him/her under my/our guidance and supervision.. This report in any form has not been submitted to any other University or Institute for any purpose.

Project Coordinator

Mr. Anith Krishnan.
Assistant Professor
Dept. of Electrical & Electronics
College of Engineering Kidangoor

Head of the Department

Dr. Indhu P. Nair
Associate Professor
Dept. of Electrical & Electronics Engineering
College of Engineering Kidangoor

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JESTIN JAISON

Ringkasan

Overall electric energy conservation and improvement in electrical energy efficiency has been a topic of interest to the electric power industry for long time, since 40% of total primary energy used is in the form of electricity. There are many facets to the energy conservation and/or energy efficiency techniques. Energy conservation by voltage reduction popularly called "CVR" has recently gained some momentum, in particular, to reduce the peak demand. This is also an integral part of the discussion in the future smarter grid technology. This paper addresses the pros and cons of the Conservation Voltage Reduction application including a thorough literature search, practical applications and a quantifiable measure of its actual amount of energy saving from utilities and the cost associated with it. It is expected that this paper will serve as a foundation to all practicing engineers addressing the issue of CVR.

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ABBREVIATIONS

HAS	High Altitude Simulation
LMTD	Logarithmic Mean Temperature Difference
PDF	Probability Density Function

NOTATION

A Area, m^2
E Voltage, V
Re Reynolds number
T Temperature, K

Bab 3

INTRODUCTION

Introduction

3.1 GENERAL BACKGROUND

Bab 4

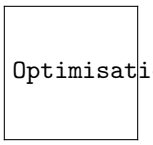
LITERATURE SURVEY

4.1 THEORETICAL INVESTIGATIONS

Theoritical Investigation

4.1.1 Optimisation studies

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Gambar 4.1: Name of the image

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4.1.2 Sample Equation

This is sample of equation 4.1

$$CVR_f = \sqrt{\frac{\%ReductionofQuantities + V_{rms(MAX)} + V^{(ref)}}{\%VoltageReduction}} \tag{4.1}$$

4.1.3 Sample Table

Sample table 4.1 based on American papers [1]

Sl.No	**Heading 1**	**Heading 2**	**Heading 3**
1	Detial 1	Detial 2	Detial 3
2	Detial 4	Detial 5	Detial 6
3	Detial 7	Detial 8	Detial 9

Tabel 4.1: Sample table

Bab 5

MODELLING

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RESULTS AND DISCUSSION

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Conclusion

Bibliografi

[1] Wikipedia

[2] Google.com